



Data-Driven Analysis and Predictive Modeling of Graduate Employment and Salary Outcomes in Singapore

Final Presentation

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Executive Summary

- **Objective:** Build a reproducible analytics framework to explain, classify, and segment Singapore graduate employment and salary outcomes at the program-year level.
- **Data:** Official Graduate Employment Survey records across NTU, NUS, SIT, SMU, SUSS, and SUTD.
- **Problem importance:** Salary alone misses employment stability, salary spread, university context, and career-value differences.
- **Feature engineering:** Created salary IQR, employment premium, salary stability, and CVI to capture outcome quality beyond raw salary and employment fields.
- **Key results:** Salary increased from 3,100 to 4,400; 2020 full-time permanent employment declined to 70.50%; three interpretable clusters were identified.
- **Business value:** Decision supports for students, universities, policymakers, and analysts with reproducible, dashboard-ready decision evidence.

1,550
program-year
records

2013–2024
GES timeframe

$R^2 = 0.58$
salary regression

$F1 = 0.71$
high-salary
model

$\rho = 0.98\text{--}1.00$
CVI robustness



Gap Analysis

Gist of Literature Review

- Graduate outcome studies often focus on salary or employment separately.
- Employability literature shows that graduate outcomes are multidimensional.
- Statistical learning literature supports using separate methods for explanation, prediction, and segmentation.

Why the problem matters

- Salary alone gives an incomplete view of graduate outcomes.
- Stable employment, salary spread, and program context also matter.
- Stakeholders need a reproducible decision-support framework, not only descriptive reporting.

Project response:

This project addresses the gap using trend analysis, statistical testing, regression, leakage-controlled classification, K-Means clustering, and a validated Career Value Index.



Research Questions, Methods, and Test Results

RQ	Focus	Method / Hypothesis Test	Key Result
How did salary and employment outcome change over time?	Trends over time	Descriptive EDA by year	Salary 3,100 → 4,400; FT permanent employment dipped to 70.50% in 2020
Do outcomes differ across universities?	University differences	Kruskal-Wallis + Holm-adjusted pairwise tests	All tested outcomes $p < .001$; effect sizes modest ($\epsilon^2 = 0.047-0.067$)
What factors are linked to salary and CVI?	Salary and CVI drivers	Regression + correlation validation	Salary model $R^2 = 0.58$; CVI strongly linked to employment variables
Can models predict high-outcome programs?	High-outcome prediction	Leakage-controlled classification	High-salary model strongest: $F1 = 0.71$, $ROC\ AUC = 0.91$
Can programs be grouped into useful profiles?	Program profiles	K-Means + PCA visualization	$k = 3$ clusters; silhouette = 0.32; interpretable profiles

Each research question was matched with an appropriate statistical or machine learning method to test trends, differences, associations, prediction, and segmentation.

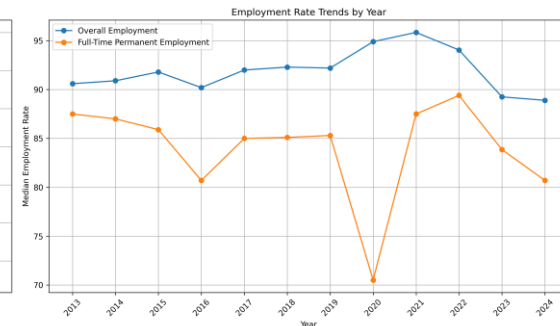
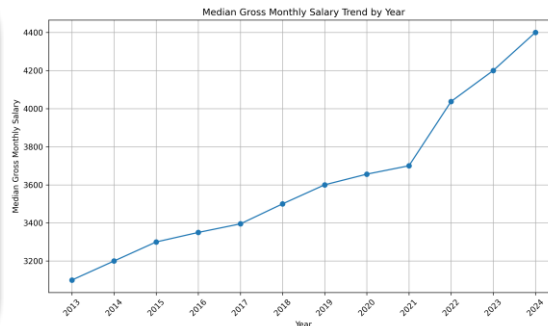
Data Description and EDA

Dataset and preparation

- Official GES data, 2013–2024
- 1,550 records; 1,445 complete cases
- 6 universities, 78 schools, 419 degree programs
- Unit of analysis: university-degree-year / program-year record
- 105 key-outcome missing records retained; no imputation

EDA interpretation

- Median gross monthly salary increased from 3,100 in 2013 to 4,400 in 2024.
- Overall employment remained generally high.
- Full-time permanent employment varied more strongly and dropped sharply in 2020.
- This supports interpreting salary and stable employment separately.





Architecture diagram/Workflow



Legend: Blue boxes = data preparation and analysis steps | Green box = final decision-support outputs | Arrows = reproducible workflow sequence

- **Pipeline purpose:** convert official program-level GES data into reproducible evidence for trend analysis, university comparison, salary modeling, high-outcome prediction, clustering, and CVI-based interpretation.
- **Output value:** notebook-generated tables and figures can be refreshed when new GES data is released and reused for dashboard-based decision support.

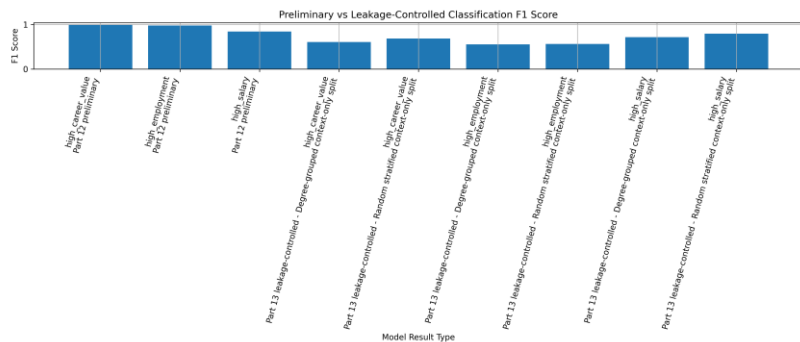


Model building and Justification

Component	Choice	Why this choice fits the problem
Statistics	Kruskal-Wallis + Holm-adjusted pairwise tests	Compares university-level outcomes without assuming normal distributions
Regression	Linear regression, 75:25 split	Explains factors associated with median salary using MAE, RMSE, and R^2 .
Classification	Logistic Regression, Decision Tree, Random Forest	Compares high-outcome prediction; final claim uses leakage-controlled context-only features.
Clustering	K-Means + PCA	Groups programs into interpretable employability-salary profiles.
CVI	Correlation + sensitivity analysis	Tests whether CVI represents meaningful outcomes and remains stable under alternative weights.

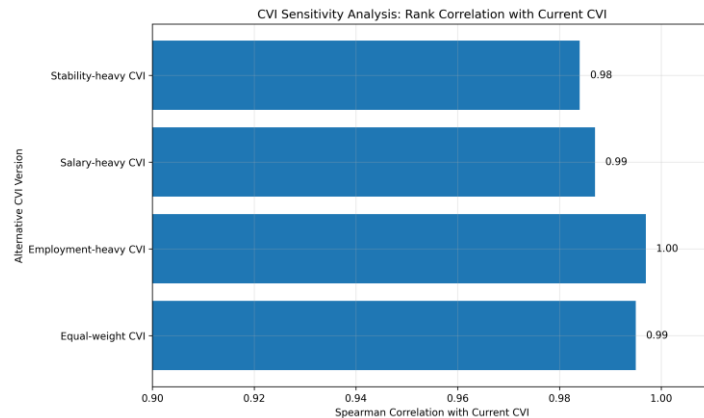
- Deep learning and agentic models were not used because the dataset is structured, tabular, and aggregated at the program-year level.
- The project prioritizes interpretability, statistical transparency, and reproducibility.

Results: Model Evidence and Visual Inferences



Classification inference

- Leakage-controlled validation gives the most defensible prediction result.
- High-salary classification was strongest: F1 = 0.71, ROC AUC = 0.91.



CVI inference

- CVI remained stable under equal-weight and alternative weighting schemes.
- Spearman ρ = 0.98–1.00; top-quartile overlap = 90.06%–95.86%.



Business Value and Stakeholder Benefits

Target user	How the system is used	Benefit
Students	Compare salary, FT permanent employment, salary spread, CVI, and cluster profile.	Better program selection beyond headline salary.
Universities	Monitor programs by cluster profile and CVI trend.	Identify programs needing career guidance or employer engagement.
Policymakers	Refresh the workflow when new GES data is released.	Track labor-market alignment and outcome changes over time.
Analysts / evaluators	Use GitHub artifacts and notebook outputs.	Reproduce tables, figures, and model results.

- **Deployment concept:** A reproducible Python/Jupyter notebook that generates tables and figures, which can be extended to Tableau, Power BI, or a web dashboard.
- **Example use case:** A student or administrator compares two degree programs using salary, stable employment, salary IQR, CVI, and cluster profile.



Conclusions, Limitations, and Future Work

Conclusions

- Graduate outcomes should be interpreted as multidimensional program-level evidence, not salary-only results.
- Salary increased from 2013 to 2024, but stable employment varied more across years.
- University differences were statistically significant but modest.
- High-salary classification was the strongest predictive target after leakage control.
- CVI was statistically validated and robust under alternative weighting schemes.

Responsible Use and Future Scope

- **Limitations:** program-year data only, observational analysis, complete-case modelling, class imbalance, and descriptive cluster interpretation.
- No individual graduate prediction and no causal claims are made.
- **Future work:** time-based validation, new annual GES updates, interactive dashboard, adjustable CVI weights, and external labor-market indicators



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Thank you